STUDENT TEAM WEEKLY REPORT: MODULE 6

PROJECT / FEATURE	TEAM & MENTOR	WEEK / LEADER
Question Bank & Quiz Engine [Auto-Grading Engine, Subjective Evaluation & Result Management]	**TEAM ID:** 3 **FACULTY MENTOR:** Mr. Rudragouda Patil	**WEEK ENDING:** 21/09/2026 **STUDENT LEADER:** Sumegh Bhamre

WORK COMPLETED	CHALLENGES / RISKS	HELP NEEDED
• **Objective Auto-Grading Engine:** Developed O(1) hash table evaluation comparing student answers against master keys for MCQ and True/False questions in QuizAttemptServiceImpl.java. • **Comprehensive Score Metrics:** Implemented automated calculation of total score, correct count, incorrect count, skipped count, unattempted count, score percentage, and pass/fail status (60% threshold). • **Subjective Manual Grading Endpoint:** Created POST /api/quiz-results/grade (QuizResultGradeHandler) allowing instructors/admins to award marks and feedback on descriptive answers. • **Two-Tier Grading Status:** Introduced AWAITING_EVALUATION and GRADED states to handle hybrid assessments with pending descriptive questions. • **Admin Results Filtering & Export:** Implemented query parameter filtering (?quizId=...&studentId=...&status=...) and CSV spreadsheet export engine.	• **False Negative Auto-Grading:** Extra whitespace or casing in True/False questions caused incorrect grading. Action: Standardized answer evaluation with .trim().equalsIgnoreCase(). **GREEN (RESOLVED)** • **Premature "Fail" Tagging:** Hybrid quizzes with subjective answers showed "Failed" before faculty graded them. Action: Added PENDING_GRADING and AWAITING_EVALUATION statuses. **GREEN (RESOLVED)** • **Subjective Score Clamping:** Evaluators could accidentally enter marks higher than maxMarks. Action: Added backend mark boundary clamping (0 <= awarded <= maxMarks). **GREEN (RESOLVED)**	**Admin / Faculty UI Verification:** Request the Frontend team to test the manual grading modal on admin.html and author.html against descriptive submissions to ensure feedback textareas serialize correctly.

INTEGRATION AND COLLABORATION
Frontend Team: Bound instant scorecard modal (#scorecardModal) and attempt review modal (openAttemptReview) to backend result fields.
Database Team: Verified PostgreSQL quiz_attempt and student_answer schema updates for subjective marks and evaluator feedback.
Integration Team: Validated end-to-end flow from student submission → auto-grade → faculty review → final scorecard.
Pipeline Learnings: Gained deep architectural insight into automated assessment pipelines and stateful grading lifecycles.
COMMON KPI EVIDENCES
Instant Grading Latency: Objective auto-grading completed in <10 ms per quiz submission.
Grading Accuracy: 100% grading accuracy verified across 50 sample MCQ and True/False test scenarios.
Export Integrity: CSV export correctly formatted with RFC 4180 quotation escaping across all attempt records.

NEXT WEEK COMMITMENTS	MENTOR RESPONSE
• **Sumegh Bhamre:** Design psychometric item analysis algorithm (Classical Test Theory Discrimination Index D). (Due: Sep 24, 2026) • **Sumegh Bhamre:** Implement multi-dimensional reporting API supporting student, class, subject, and quiz breakdowns. (Due: Sep 26, 2026)	